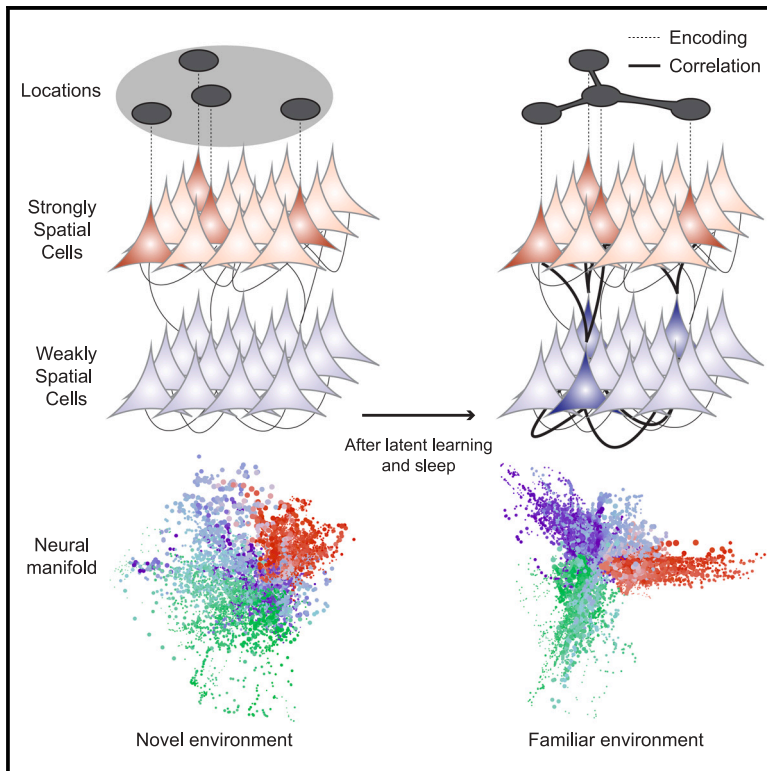


Latent learning drives sleep-dependent plasticity in distinct CA1 subpopulations

Graphical abstract



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In brief

Guo et al. find that when mice repeatedly explore a novel environment, a distinct subset of neurons in the CA1 region of the hippocampus, which initially exhibits weak spatial selectivity, gradually develops correlated activity with place cells, establishing a map-like ensemble code that represents the animal's spatial experience.

Highlights

- The CA1 neural ensemble forms a low-dimensional manifold during latent learning
- A subset of weakly spatial CA1 neurons increases correlated activity with other neurons
- The plasticity of the CA1 ensemble following latent learning is linked to sleep



Article

Latent learning drives sleep-dependent plasticity in distinct CA1 subpopulations

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SUMMARY

Latent learning is a process that enables the brain to transform experiences into “cognitive maps,” a form of implicit memory, without requiring reinforced training. To investigate its neural mechanisms, we record from hippocampal neurons in mice during latent learning of spatial maps and observe that the high-dimensional neural state space gradually transforms into a low-dimensional manifold that closely resembles the physical environment. This transformation process is associated with the neural reactivation of navigational experiences during sleep. Additionally, we identify a subset of hippocampal neurons that, rather than forming place fields in a novel environment, maintain weak spatial tuning but gradually develop correlated activity with other neurons. The elevated correlation introduces redundancy into the ensemble code, transforming the neural state space into a low-dimensional manifold that effectively links discrete place fields of place cells into a map-like structure. These results suggest a potential mechanism for latent learning of spatial maps in the hippocampus.

INTRODUCTION

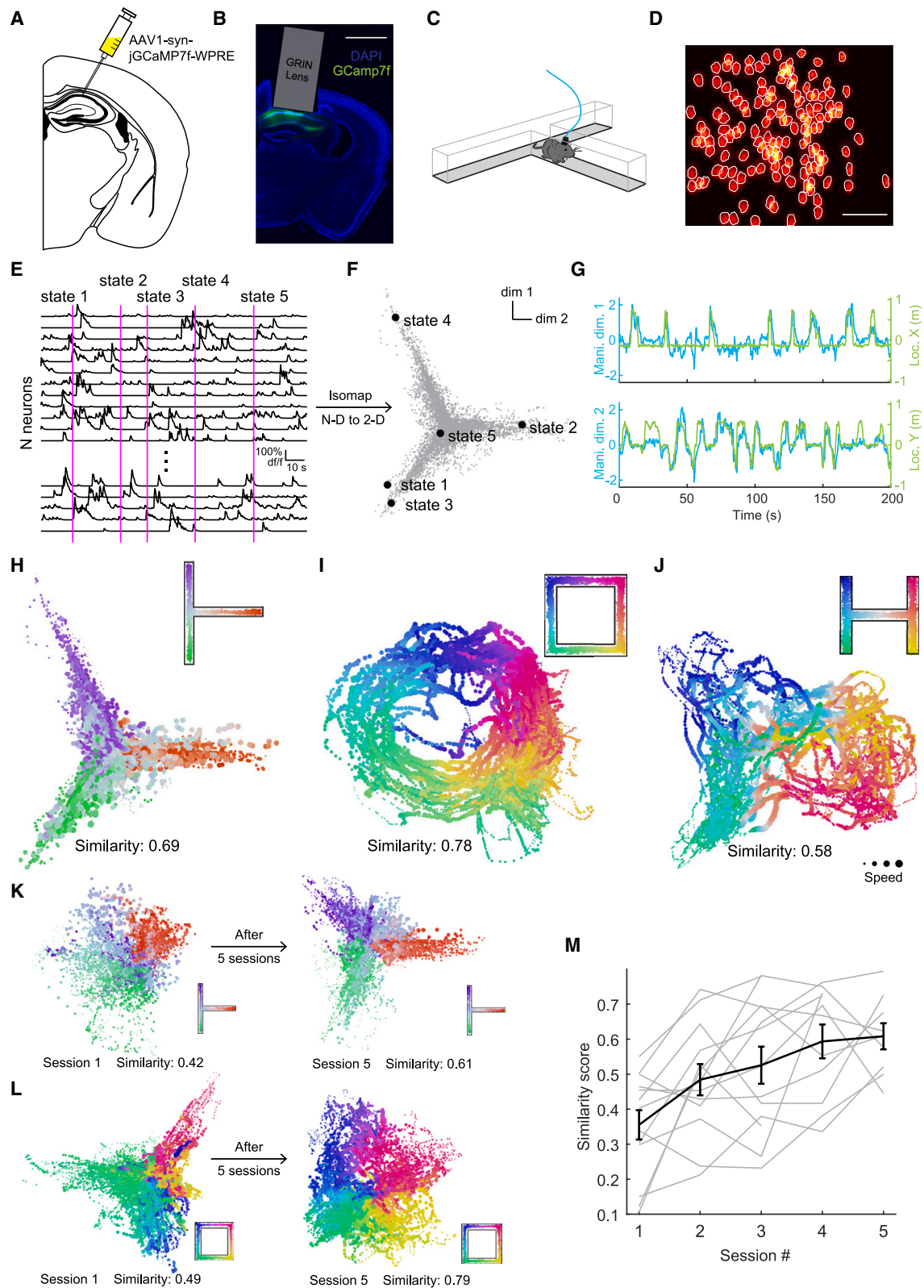
In his seminal work from 1948, Edward Tolman introduced the “cognitive map” theory, postulating that the brain organizes past experiences into a memory structure resembling a map, serving as a guiding framework for future actions.^{1–3} This theory gained substantial support with the discovery of hippocampal place cells, characterized by their distinct spatial tuning functions, suggesting a potential neural substrate for constructing mental representations of the environment.^{4,5} However, previous studies have primarily focused on subjects engaged in foraging or reinforcement learning tasks, limiting our observations to neural activities associated with overt behavioral changes. Nonetheless, cognitive maps can be acquired and stored implicitly. For instance, Tolman’s original work introduced the concept of latent learning, where rats could surreptitiously acquire knowledge of a maze’s layout by freely exploring it for a few days without the need for reinforced training or displaying discernible behavioral changes.^{1,6} Subsequent studies demonstrated that this process also requires sleep.⁷ While it is well documented that place cells form their spatial tuning rapidly—usually within minutes—and remain stable for days,^{8,9} what the hippocampal circuit learns during latent learning and sleep remains poorly understood.^{2,10–12}

However, hippocampal neurons do not operate independently in encoding spatial information (I_p); rather, they function collectively as an ensemble, with individual neurons exhibiting

temporally correlated activity during active navigation and sleep.^{13–16} Consequently, while the impact of latent learning on individual neurons may be subtle, its effects may manifest more prominently within the hippocampal neural ensemble and the correlated firing patterns among neurons, potentially altering the population code without significantly affecting individual neurons.^{17,18} To observe such effects, it is imperative to monitor changes in hippocampal neurons as an ensemble during latent learning and sleep, with a focus on interneuronal relationships.

In this study, we recorded the activity of hundreds of neurons within the CA1 region of mice over the course of several days during maze-based latent learning and investigated changes within the neural ensemble during both maze runs and sleep. We treated instantaneous population activity patterns of neurons as high-dimensional neural states, with the collective data from a recording session serving as a state space. Using manifold learning techniques, we transformed the high-dimensional neural state space into low-dimensional representations, revealing substantial changes in the population code following latent learning and post-learning sleep. Moreover, we identified a distinct subset of CA1 neurons characterized by limited spatial tuning and subtle changes during latent learning, yet these neurons played a pivotal part in driving the changes in the ensemble code. Collectively, these findings expand our understanding of the hippocampal neural code and spatial memory, particularly the formation of cognitive maps during latent learning.





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RESULTS

Latent learning reorganizes the hippocampal neural state space into manifolds resembling navigational experiences

We expressed Ca^{2+} indicator jRCaMP7f¹⁹ in the dorsal hippocampus of mice. Using a miniature head-mounted single-photon microscope,^{20,21} we recorded Ca^{2+} activity of neurons in the pyramidal layer of the CA1 region through a chronically implanted gradient-index (GRIN) lens. The mice were assigned to mazes with varying geometric shapes and given daily free exploration runs for 30–40 min over 5–7 days, with their activity monitored.²² The mazes were devoid of any reward or punishment associations (Figures 1A–1C and S1). Regions of interest (ROIs) were extracted^{23,24} and treated presumptively as neurons, with their Ca^{2+} levels reflecting neural activity (Figure 1E). The collective Ca^{2+} activity during a session was regarded as an N -dimensional point cloud, where N represents the number of neurons and each point's location is determined by the population neural firing rates within a 100-ms window. Consequently, each point represents the ensemble neural state—a snapshot of all the neurons at a given time (Figure 1E).

Recent studies have demonstrated that the intrinsic dynamics of neural ensembles often reside on a low-dimensional manifold within the broader high-dimensional state space.^{25–28} This can be visualized after proper dimensionality reduction transformations. We employed Isomap,²⁹ an unsupervised manifold learning algorithm, to establish a low-dimensional representation of the high-dimensional state space. The resulting manifold retains the geodesic distance between all pairs of neural states, assuming transitions consist primarily of a series of hops to nearby states, which mirrors the smooth state transitions observed in the hippocampus during spatial navigation.^{15,30–33} To optimize the algorithm, adaptive parameter selection was performed for each recording session to minimize information loss and prevent overfitting (Figure S2). Because of the simple geometric shapes of our mazes (Figure S1), we transformed all recording sessions into two-dimensional (2D) manifolds while acknowledging that in many instances, additional dimensions contain information not present in the first two (Figure S3).

Our analysis revealed that when the animal became familiar with the maze after 3–5 days of exploration sessions, its spatial experiences were intrinsically encoded within the neural manifold (Figures 1F and 1G). In other words, without any information regarding the animal's behavior or the spatial tuning of individual

neurons, the 2D manifold of neural states revealed by Isomap closely resembled the animal's physical occupancy in the maze (Figures 1F and 1H). Furthermore, we defined a “mental trajectory” comprised of neural state transitions on the 2D manifold that could be compared with the animal's actual path in the maze (Figure 1G). This similarity was quantifiable through a similarity score calculated as the average correlation coefficient between the mental and physical trajectory (see STAR Methods). We observed high similarity scores across multiple animals navigating mazes of various shapes, underscoring the faithful representation of spatial maps within the internal states of the hippocampus (Figures 1H–1J). Notably, the neural manifolds exhibited a topological encoding of maze structures, transforming a T maze into a Y shape and a square loop maze into a circular pattern. These findings align with previous research positing topological cognitive maps within the hippocampus.^{34–36}

However, when animals were first introduced to a novel maze, the similarity between the neural manifold and the physical maze was notably low (Figures 1K and 1L). As the animal engaged in repeated exploration of the maze over several sessions, the similarity score progressively improved, as reflected in an increased resemblance between the manifold and the maze layout (Figures 1K–1M). Notably, this enhancement in similarity was not accompanied by significant alterations in the animal's exploratory behavior, including metrics like distance traveled, movement speed, or acceleration (Figure S4). In sum, after several days of latent learning, the neural manifold gradually transformed into a structure that intrinsically encodes the spatial environment, closely resembling the cognitive map formation process proposed by Tolman.¹

The formation of the hippocampal neural manifold depends on sleep

It is well established that sleep plays an important role in learning and memory consolidation, but the specific processes in the hippocampus during sleep that contribute to neural plasticity remain elusive.³⁷ In particular, sleep is known to play a significant role during latent learning,⁷ yet its underlying mechanism remains unclear. We hypothesize that by analyzing changes within the neural manifold, we can identify the contributions of sleep following learning.

To test this hypothesis, we exposed mice to a novel maze for two separate free exploration sessions within the same day. Between these runs, the animals spent 3 h in their home cages (Figures 2A–2D). Half of the mice (sleep group, Figures 2A and

Figure 1. Latent learning induces plasticity in neural manifold representation of space in CA1

- (A) Virus delivery of transgenes.
(B) Coronal section of a mouse brain showing jRCaMP7f expression (green) and GRIN lens location. Scale bar: 1 mm.
(C) Recording from a freely exploring mouse.
(D) Example footprints of isolated ROIs. Scale bar: 0.1 mm.
(E) Example segment of Ca^{2+} traces of hippocampal neurons.
(F) Example 2D manifold calculated from data in (E). The 2D states correspond to states labeled in (E).
(G) Comparison of the neural state trajectory on the manifold and the animal's spatial trajectory.
(H–J) Example manifold representations of 3 mice running 3 different mazes. Each dot corresponds to a populational neural state during a 100-ms window, with the size of the dot representing the speed of the animal's movement (range: 0–1 m/s), and the color of the dot representing the animal's location in the maze (inset).
(K and L) Example manifold representations of a maze early and late during latent learning.
(M) Group data showing the improvement of similarity score throughout latent learning ($F_4 = 5.54$, $p = 0.002$, repeated measures ANOVA. 13 mazes [5 circles, 3 T's, 3 H's, 2 I's], black curve shows mean and SEM from 10 mice).

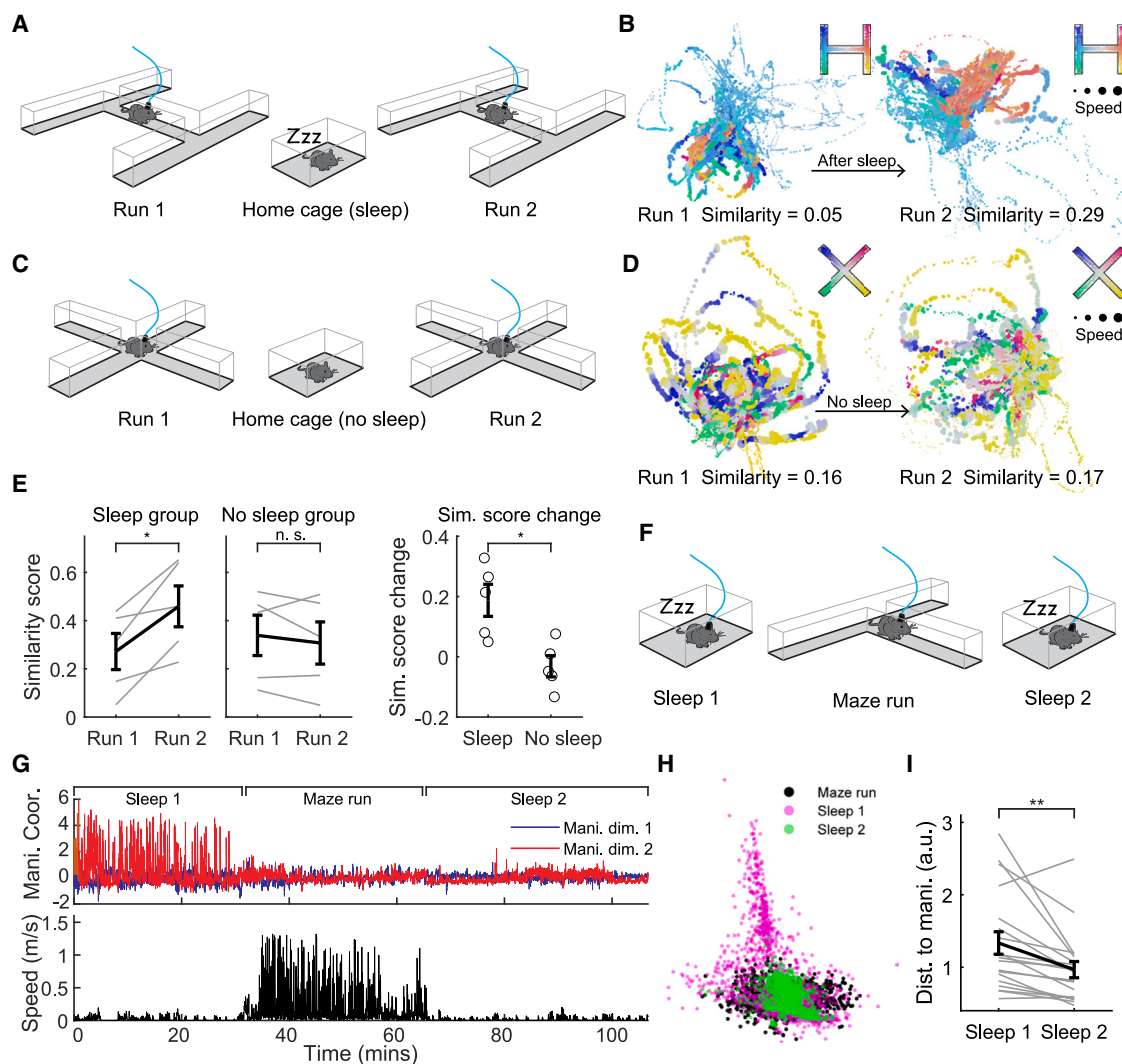


Figure 2. Neural manifold plasticity is sleep dependent

(A and C) Cartoons showing the task. The mice are either allowed to or prevented from sleep during a 3-h break between two maze runs.

(B and D) Example neural manifolds from the sleep group (B) and the no-sleep group (D).

(E) Similarity score significantly improved in the sleep group ($T_4 = 3.51$, $p = 0.02$, paired Student's *t* test, 5 sessions, 5 mice, 2 H's and 3 X's) but not in the no-sleep group ($T_4 = 0.9$, $p = 0.42$, paired Student's *t* test, 5 sessions, 5 mice, 3 H's and 2 X's). There is a significant difference in the similarity score change between groups ($T_4 = 3.42$, $p = 0.009$, unpaired Student's *t* test, 5 sessions, 5 mice) (between runs, $F_1 = 0.85$, $p = 0.37$; between groups: $F_1 = 0.25$, $p = 0.62$, two-way ANOVA). Black curves show mean and SEM.

(F) Cartoon showing the task investigating neural activity during sleep before and after a maze run.

(G) Top: example trajectory of neural states on the manifold constructed from maze run data. Bottom: animal's speed. Mani. Coord., manifold coordinates.

(H) 2D representation of the manifold in (G).

(I) Neural states during the pre-sun sleep are further away from the manifold than the post-run sleep ($F_1 = 3.15$, $p = 0.023$, repeated measures ANOVA, black curve shows mean and SEM from 19 sessions, 4 mice).

* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

2B) were permitted to sleep, while the remainder (no-sleep group, Figures 2C and 2D) were prevented from doing so. Animals in the sleep group showed significantly more sleep-like or immobilized states than those in the no-sleep group (Figures S5A–S5C). Comparing the neural manifolds from the two maze runs, we observed a significant improvement in the similarity score for animals allowed to sleep, with no change

evident in those deprived of sleep (Figure 2E). These findings suggest that changes in the hippocampal neural manifold are contingent on sleep. However, such differences could potentially be caused by stress or behavioral changes induced by sleep deprivation. If neural manifold changes were indeed linked to sleep, then we would expect to find evidence in neural activity during sleep.

Subsequently, we recorded hippocampal activity during a maze run session and two sleep sessions—one immediately before and one after the maze run (Figures 2F, S5D, and S5E). The neural data during the maze run session were transformed into 2D representations using Isomap, and the transform was applied to the sleep data, effectively “decoding” neural states during sleep as point clouds in the same subspace as the maze run manifold (Figure 2G). Comparing the two sleep sessions, we observed that post-run sleep data were located significantly closer to the maze run manifold (Figures 2G–2I). This suggests that during post-run sleep, the hippocampal neurons were preferentially exhibiting ensemble activity similar to that observed during the run, consistent with previous work describing the hippocampal replay phenomenon, which is crucial for memory consolidation within the hippocampal-cortical circuit.^{14,38–40} In summary, the neural manifold changes that improve similarity to the mazes are associated with post-maze-run sleep and reactivation of neural activity during sleep.

Hippocampal neural ensembles show improved spatial and mental tuning after latent learning and sleep

Next, we examined the underlying factors driving changes in the neural manifold during latent learning. For each neuron, we calculated its “place fields,” i.e., specific spatial locations that drive their firing activity, and quantified its tuning quality using I_p , a mutual information measure between locations and neurons’ firing rates.⁴¹ We observed a wide spectrum of I_p values, signifying varying degrees of specificity in the neurons’ responses to spatial locations (Figure 3A).

In addition to the place fields, we applied a similar analysis to the neural manifold, which we termed “mental fields” for each neuron. Instead of associating neural activity with the animal’s physical trajectory in the maze, this analysis associates each neuron’s firing rate with a mental trajectory on the manifold. Conceptually, this measurement describes a neuron’s tendency to fire with respect to particular ensemble neural states, i.e., the activity of other neurons. Similar to the place fields, we quantified the quality of mental fields as mental information (I_m) and observed a broad range of I_m values in hippocampal neurons (Figure 3B).

We conducted a comparison of the I_p and I_m values of hippocampal neurons throughout the latent learning process. While the distributions over time exhibited considerable overlap (Figures 3C and 3D), the average values of both I_p and I_m showed a significant increase following five sessions of free exploration (Figure 3E). This suggests that subtle tuning shifts may occur in a select group of neurons during latent learning.

We further explored the role of sleep in changing neuronal spatial and mental tuning following latent learning. By comparing I_p and I_m values of the same neurons in two maze runs separated by 3 h of home cage rest, with or without sleep, we found that sleep significantly enhanced both the I_p and I_m values (Figure 4). These findings reinforce our earlier results, indicating that sleep plays a crucial role in hippocampal functions during latent learning.

A subset of weakly spatial neurons in the hippocampus undergo tuning changes following latent learning

In most studies of the hippocampus, neurons are first classified as place cells or “non-place cells” according to their spatial tun-

ing quality, with further analyses often only conducted on the place cells, which comprise less than half of the recorded population. It is well established that in a novel environment, place cells promptly establish their place fields, typically within a matter of minutes, and maintain relative stability for several days.^{8,9} However, this phenomenon appears to be at odds with our observations, which indicate an enhancement in ensemble place coding over the course of latent learning. To account for this apparent discrepancy, we hypothesized that the often-neglected non-place cells undergo changes that could explain the shifts in the neural manifolds.

To test this hypothesis, we used an algorithm optimized for Ca^{2+} data to identify neurons that were consistently tracked for at least five consecutive sessions.⁴² Within this subset of tracked neurons, we identified what can be termed classic place cells—neurons that displayed distinct place fields during the initial session and maintained stability in subsequent sessions (Figure 5A). The mental fields of these place cells also exhibited a high degree of specificity from the first session and continued to do so throughout the learning process. Consequently, both the I_p and I_m for these neurons remained consistently high and displayed no significant changes over time (Figure 5B). However, we also observed a subset of neurons with weak spatial tuning during the first session (Figure 5C). Interestingly, these neurons displayed subtle but discernible improvements in both their I_p and I_m values over the course of latent learning (Figure 5D). Notably, even after five learning sessions, the spatial tuning quality of these neurons remained lower than that of the place cells. Moreover, these neurons showed increasing specificity in the mental fields as the I_m increased, suggesting that, instead of becoming more sensitive to a specific location, these neurons were becoming more sensitive to a specific ensemble state—that is, whenever a specific set of neurons is active.

To characterize the dynamics of these tuning changes, we calculated two “growth slopes” for each of the tracked neurons, one for I_p and one for I_m , assessing the patterns of change over the course of latent learning (Figures 5B and 5D). We then explored whether these growth slopes were linked to the initial tuning quality of neurons during the first session. Interestingly, our analysis revealed that neurons with lower initial place information exhibited a higher rate of growth in both I_p and I_m (Figure 5E). We categorized the tracked neurons into two groups based on their initial place information: weakly spatial cells (WS cells; with initial I_p values below 0.4) and strongly spatial cells (SS cells; with initial I_p values above 0.4). We observed that WS cells displayed significantly more growth in both I_p and I_m compared to SS cells (Figure 5F). Notably, for SS cells, the average I_p growth slope was centered at zero, aligning with prior research that has described a stable set of place cells.^{8,9} Indeed, the average growth curves demonstrated that SS cells exhibited no significant changes in either I_p or I_m , while WS cells experienced clear growth over five sessions (Figure 5G). We also calculated changes in the sparsity of neurons’ place and mental fields. Similarly, SS cells showed little change in sparsity, while WS cells showed a clear decrease of sparsity after latent learning (Figure S6A). We performed similar analyses with varying thresholds to divide the cell population (0.3 and 0.5 on the initial I_p) and showed that the results were not qualitatively altered

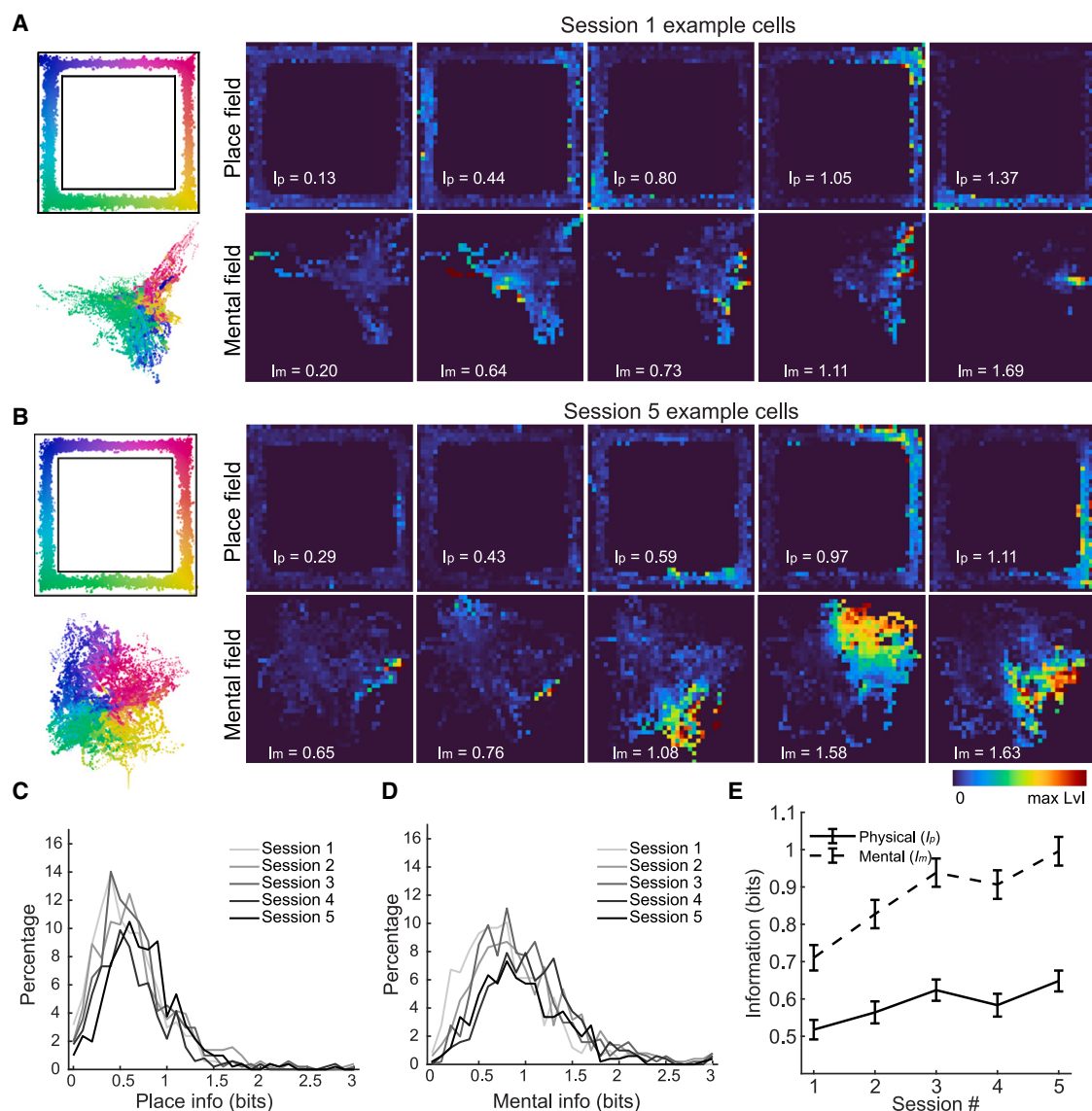


Figure 3. Latent learning induces heterogeneous neural tuning changes in CA1 neurons

(A) Example neurons' place field tuning (top row) and mental field tuning (bottom row) from an early exploration session. Tuning quality ranges from low to high (left to right).

(B) Similar to (A) but with neurons from the same animal during a late exploration session.

(C) Distributions of neurons' place information across multiple latent learning sessions.

(D) Distributions of neurons' mental information across multiple latent learning sessions.

(E) Comparisons of the average information changes from session 1 to session 5. Average place information of the neural ensemble significantly improved ($F_4 = 11.34, p = 41.12 \times 10^{-9}$, one-way ANOVA). Average mental information of the neural ensemble also significantly improved ($F_4 = 22.81, p = 4.12 \times 10^{-9}$, one-way ANOVA) (curves show mean and SEM, $n = 2,045, 1,766, 1,627, 1,662$, and $1,522$ for each session pooled from 10 mice).

(Figures S6D and S6E vs. Figures 5G, S6G, and S6H). Additionally, we divided the cell population based on the averaged I_p during all 5 sessions (using the modal value 0.5 of the distribution as the new dividing point) and showed qualitatively similar results (Figures S6F and S6I). These analyses suggest that SS and WS cell groups might be evolving with distinct paths during latent learning. Indeed, we noticed a plateauing effect of WS cells toward later sessions (Figure 5G, main effect for time not statistically significant under repeated measures ANOVA tests),

which suggests that WS cells may not improve infinitely during learning.

We systematically compared the putative SS and WS cell groups to see if there were any functional differences (Figure S7). We observed that SS and WS cells were all recorded and tracked with similar quality (Figures S7A–S7E), with their peak, averaged, signal-to-noise, and signal variability comparable (Figures S7F–S7I). The only major difference was the size of their place fields, with the SS cells' being significantly smaller.

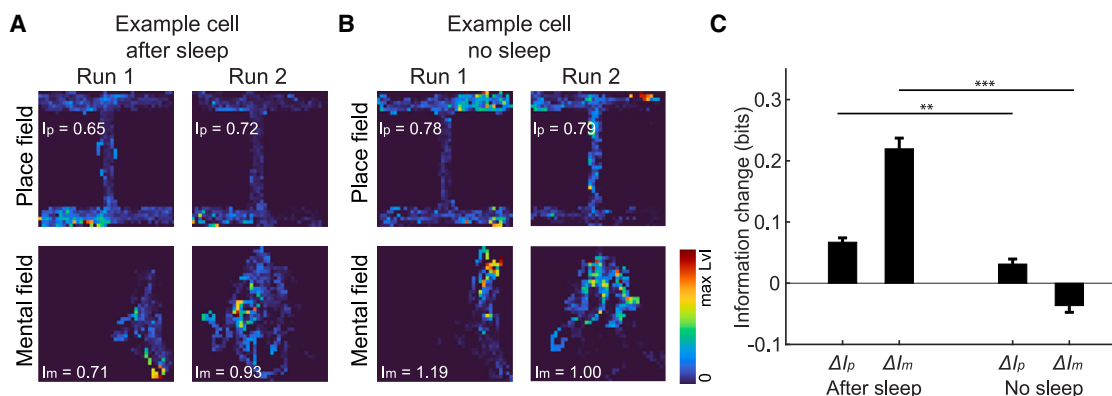


Figure 4. Improved spatial and mental tuning after sleep

(A) Top: the place fields of a neuron calculated from two runs separated by a 3-h sleep period. Bottom: mental fields from the same neuron. (B) Same analysis of an example neuron as in (A) but no sleep was allowed during the 3-h period between the two runs. (C) Average information changes across neurons and animals. Sleep significantly improves both the spatial information ($T_{1378} = 3.08$, $p = 0.0021$, unpaired Student's *t* test), and the mental information ($T_{1378} = 13.09$, $p = 5.78 \times 10^{-37}$, unpaired Student's *t* test). All changes were significantly different from zero (ΔI_p after sleep: $T_{545} = 9.54$, $p = 4.57 \times 10^{-20}$; ΔI_m after sleep: $T_{545} = 12.58$, $p = 4.84 \times 10^{-32}$; ΔI_p no sleep: $T_{833} = 3.82$, $p = 1.43 \times 10^{-4}$; ΔI_m no sleep: $T_{833} = -3.34$, $p = 8.89 \times 10^{-4}$; Student's *t* tests; bar plots show mean and SEM). ** $p < 0.01$ and *** $p < 0.001$.

Can the subtle changes we observed in WS cells account for the dramatic shifts in the shape of the neural manifold? To address this question, we recalculated the neural manifolds for each animal during the first two latent learning sessions. Consistent with our prior findings, the similarity score improved after latent learning. We then conducted the same manifold analysis separately for SS cells and WS cells. When using only SS cells, the initial similarity scores of the manifolds were comparable to the all-cells group, yet they exhibited no significant improvement after latent learning. In contrast, the WS cell manifold displayed lower initial similarity scores and showed only marginal (statistically insignificant) improvement after latent learning (Figure 6). These results suggest that SS cells, with their rapidly established spatial tuning, do not readily contribute to the formation of neural manifolds that closely mirror the environment. Furthermore, the relative stability of SS cells rules out their role in shaping the neural manifolds during latent learning. However, when SS cells were combined with WS cells, the entire neuron ensemble underwent profound changes on the manifold level. WS cells, with their subtle changes that might easily be overlooked, appear to be a crucial factor in molding the neural manifolds.

Weakly spatial neurons develop correlated activity with other neurons and contribute to neural manifold formation

Finally, we investigated why these subtle changes in the WS cells contributed to forming the neural manifolds. Conceptually, the neural state space adopts a low-dimensional manifold structure only when redundancy is present in the neural code, often as a result of correlated activity among neurons (Figure 7A).^{18,43,44} The identification of changes in neural correlations can provide insights into the mechanisms underlying manifold plasticity.

For each neuron, we defined a correlation score as its average absolute correlation with all other neurons (calculated directly from neural activity traces over time). During maze run sessions,

the population correlation score exhibited a modest increase (Figure 7B). However, when we divided neurons into SS and WS cells, we observed that the primary source of the correlation increase was the WS cells (Figure 7B). These WS cells became considerably more correlated with other cells during latent learning, most notably with SS cells, followed by other WS cells (Figure S6B). During sleep sessions between maze runs, both SS cells and WS cells showed a consistent increase in the correlation score (Figure 7B), which can be attributed to an increase in SS-WS interactions (Figure S6C). These findings elucidate our previous observations. First, as a WS cell became more correlated with other neurons, it developed reliable firing activity within non-random locations within the neural state space, resulting in greater I_m in the mental field tuning. Second, the increased correlation of WS cells with SS cells after learning and sleep introduced redundancy into the ensemble neural code, transforming the neural state space into a low-dimensional manifold. This theory also agrees with previous observations that WS cells (or non-place cells) as a collective can predict the firing rates of individual SS cells (or place cells).⁴³

In summary, our findings propose a possible mechanism for the latent learning of cognitive maps. When an animal initially encounters a new environment, a subset of neurons in the hippocampus rapidly develop preferential tuning for specific locations, giving rise to place cells (SS cells). Subsequently, during sleep, ensemble cell activities representing the spatial experience are reactivated in the hippocampus. With repeated exposures to the environment and subsequent sleep periods, previously non-selective neurons (WS cells) begin to develop correlated firing activity with specific SS cells and with each other. We hypothesize that this correlated activity from the WS cells helps to establish connections that topologically link the discrete place fields of SS cells into a map structure (Figure 7C).^{3,45} These changes in WS cells introduce redundancy to the population code, transforming the neural state space into a low-dimensional map-like manifold that reflects past experience.

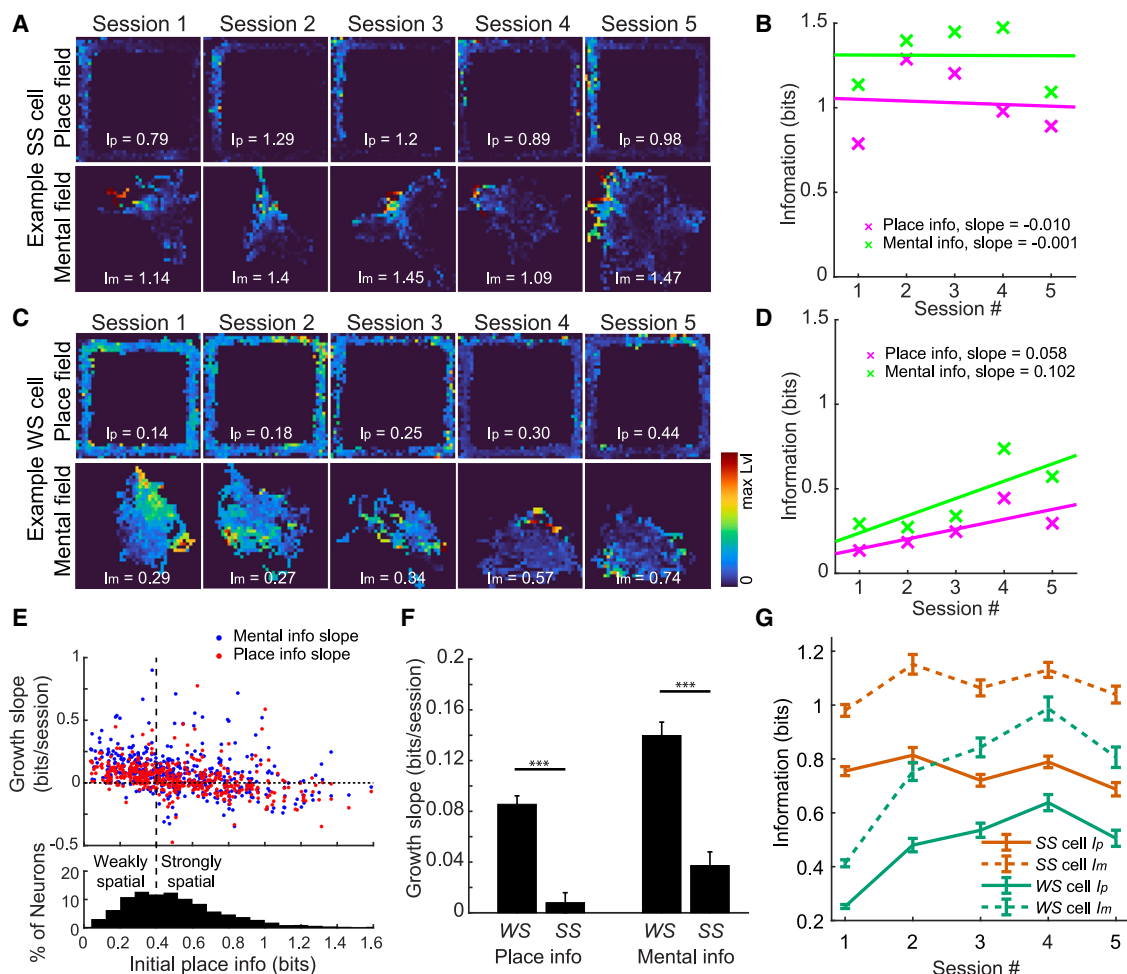


Figure 5. A subset of WS cells undergo changes during latent learning

(A) Example place and mental field tuning changes of a tracked neuron with strong spatial tuning over the course of latent learning.

(B) The growth curves of place and mental information of the neuron in (A).

(C) Same as (A) but from a tracked neuron with weak spatial tuning.

(D) Same as (B) but for the neuron from (C).

(E) Distribution of growth slopes of place and mental information (top) as a function of the neuron's initial place information (bottom). Neurons were subdivided into the strongly spatial (SS) group and weakly spatial (WS) group using 0.4 bits as a threshold.

(F) Comparison of growth slopes. For place information, WS cells had higher growth rate than SS cells ($T_{443} = 6.36$, $p = 5.10 \times 10^{-10}$, unpaired Student's t test); for mental information, WS cells had higher growth rate than SS cells ($T_{443} = 5.92$, $p = 6.57 \times 10^{-9}$, unpaired Student's t test, bar plots show mean and SEM).

(G) Information change of tracked SS and WS cells over learning (SS cells: main effect for time: $F_4 = 0.7$, $p = 0.58$; main effect for measurement type: $F_1 = 172.0$, $p = 9.49 \times 10^{-34}$, interaction: $F = 1.37$, $p = 0.24$; WS cells: main effect for time: $F_4 = 1.4$, $p = 0.25$; main effect for measurement type: $F_1 = 92.5$, $p = 1.05 \times 10^{-18}$, interaction: $F = 3.1$, $p = 0.01$; 330 SS cells, 176 WS cells from 10 mice; repeated measures ANOVA, curves show mean and SEM).

DISCUSSION

Neural manifolds and cognitive maps

Why does the hippocampus invest additional effort and employ WS cells to transform the ensemble representation into a map-like structure? This goes back to the reasoning of Tolman, who argued that maps are not only more efficient and flexible representations of experiences but also offer a generative function, which enables the brain to simulate potential actions in the environment without the need for actual physical actions. Indeed, we observed that while SS cells offer accurate representations of

the animal's current location, such information alone does not form a map-like representation. A mechanism is needed to effectively bridge these discrete locations and form a topological graph structure with locations as "nodes" connected by "edges."⁴⁵

We propose that WS cells establish these edges after latent learning and during sleep. Although not responding to specific locations like SS cells, WS cells specialize in responding to "mental locations," i.e., specific ensemble firing patterns of other cells. If a WS cell's mental field encompasses two subsets of SS cells that encode distinct locations, this WS cell can serve

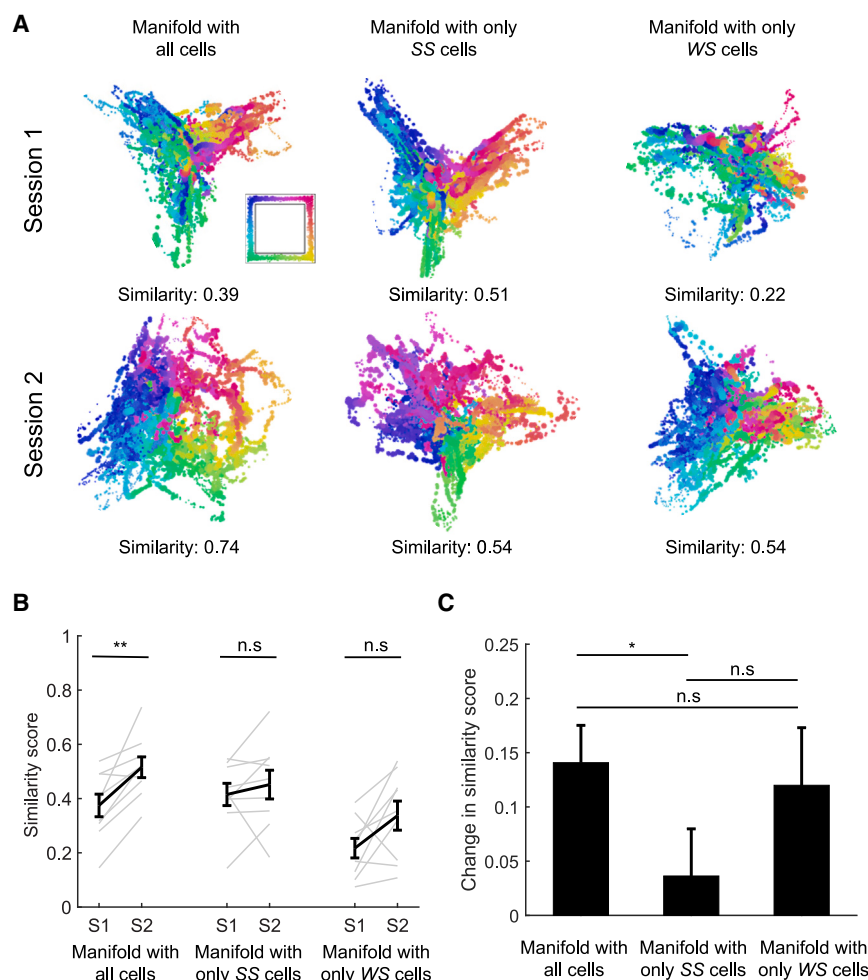


Figure 6. WS cells drive manifold plasticity after latent learning

(A) Top: neural manifolds of an animal during the first exploration session, calculated using all tracked cells, only tracked strongly spatial cells, and only tracked weakly spatial cells; Bottom: neural manifolds of the same animal from the next exploration session.

(B) Comparisons of similarity score changes across three manifold types. With all cells, the similarity score improved ($p = 0.0078$); with strongly spatial cells only, the similarity score did not change ($p = 0.25$); and with weakly spatial cells only, the similarity score marginally improved ($p = 0.053$). Curves show mean and SEM.

(C) Manifold with all cells improved significantly more than only strongly spatial neurons ($p = 0.039$). There is significant difference between other groups (SS vs. WS: $p = 0.50$, all vs. WS: $p = 0.57$). All tests were done with the Wilcoxon signed-rank test with 9 animals. Bar plots show mean and SEM. ** $p < 0.01$ and * $p < 0.05$.

as a bridge between these locations. With well-constructed SS-WS-SS-WS links, the network can potentially generate firing sequences spontaneously, without external input. Decades of studies have shown that the hippocampus indeed operates in this manner. Phenomena like theta sequences during navigation^{15,30–33} and replays during sleep and task-solving processes^{39,46–49} are all conceivable functions of a well-established cognitive map. Furthermore, numerous studies have demonstrated that the hippocampus encodes non- I_p , including sensory cues, rewards, behavioral states, and even time,^{50–53} which can be incorporated into multisensory neural manifolds that extend beyond representations of space, allowing for the storage of complex experiences and the planning of sophisticated actions.

Latent learning as a behavioral assay

Our study underscores the significance of latent learning as an assay for investigating neural functions. In the field of systems neuroscience, a prevailing approach involves initially training animals or subjects to perform specific tasks with reinforcement. For example, non-human primates are often trained to perform joystick-based tasks while maintaining eye fixation, and rodents are conditioned to navigate particular routes in mazes to obtain

rewards or respond to specific sensory cues with actions such as licking a spout or freezing. These methods offer invaluable avenues for studying the neural circuits of interest. However, a notable limitation of these approaches is that by the time recordings commence, the animals have typically acquired the ability to solve the tasks using stereotypical actions, often relying on rote memory with minimal thinking or planning. Consequently, the neural activity frequently exhibits precise and stereotypical patterns that mirror the behavior, lacking the natural variability

observed in untrained settings, which can limit our assessment of their functions. In this study, we focused on animals behaving naturally and demonstrated that during freely exploratory behavior and subsequent sleep, in the absence of reinforcement, substantial neural plastic changes at the ensemble level still occur. This form of implicit and unsupervised learning constitutes a crucial facet of human learning and intelligence, warranting further in-depth investigations.^{54–56}

Manifold learning methods in data analysis

This study also underscores the significance of manifold learning as a vital tool for analyzing high-dimensional neural datasets. With advancements in both electrophysiology and optical imaging technologies, researchers can now simultaneously monitor hundreds or even thousands of neurons. Effectively handling and analyzing such datasets has become a major endeavor in the field. Intuitively, the complexity of directly visualizing high-dimensional data necessitates dimensionality reduction methods. Consequently, numerous studies have employed a wide array of techniques to transform their recordings into lower dimensions for further analysis.⁵⁷ Recently, non-linear manifold learning methods such as t-distributed stochastic neighbor

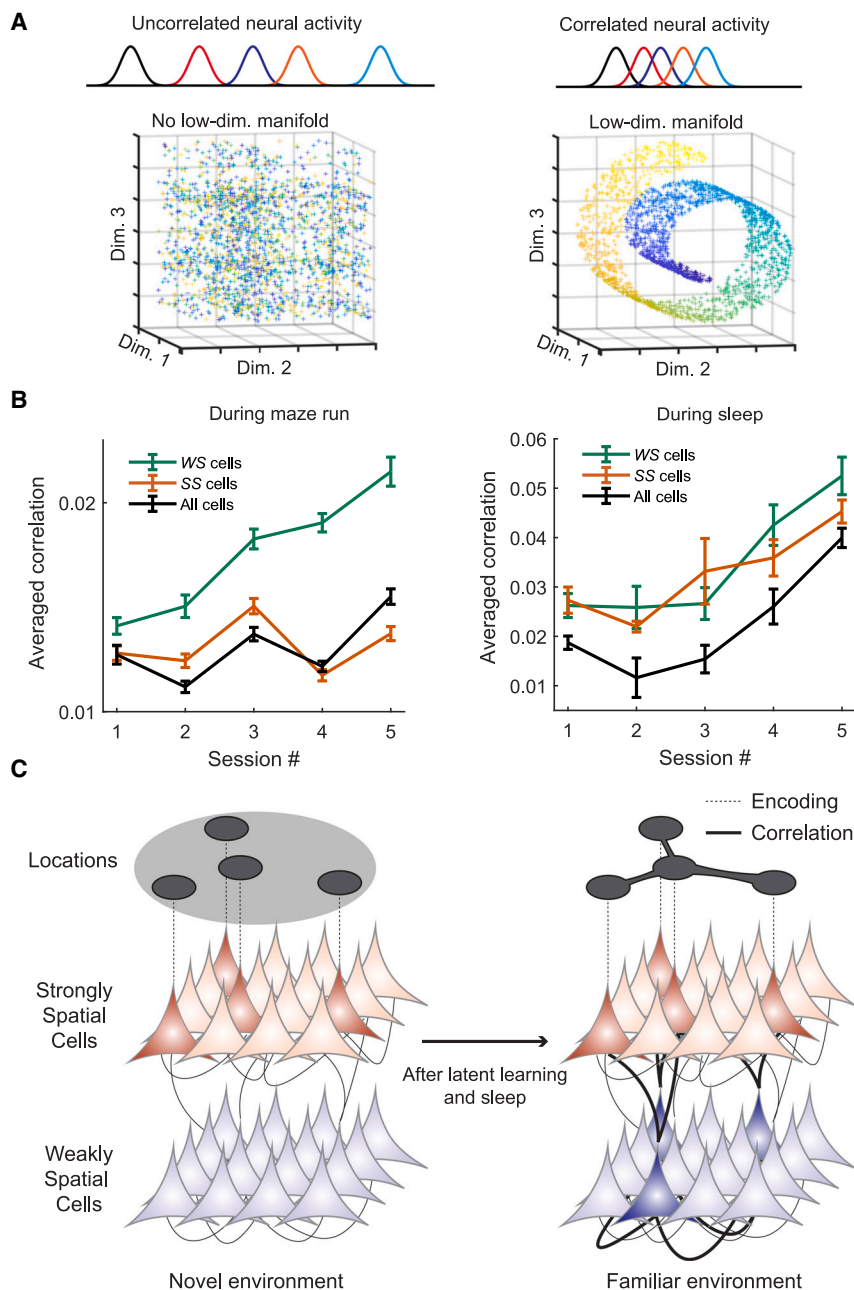


Figure 7. Latent learning elevates neural correlation in weakly spatial cells

(A) Cartoons showing relationship between neuronal correlation and neural manifolds.

(B) Averaged correlation changes over learning (during maze run: main effect for time: $F_4 = 20.6$, $p = 6.48 \times 10^{-17}$, main effect for cell type: $F_2 = 290.55$, $p = 1.81 \times 10^{-64}$; main effect two-way ANOVA, 330 SS cells, 176 WS cells from 10 mice; during sleep: main effect for time: $F_4 = 24.6$, $p = 9.30 \times 10^{-11}$, main effect for cell type: $F_2 = 1.27$, $p = 0.29$; main effect two-way ANOVA, 257 SS cells, 216 WS cells from 4 mice; curves show mean and SEM).

(C) Cartoon showing the conceptual model for latent learning.

we were not only able to identify a distinct subset of neurons that had often been overlooked by previous studies but were also able to interpret manifold plasticity as changes in neural correlation and code redundancy,^{18,43} providing a much-needed biological foundation to the mathematical description of our data.

Limitations of the study

A major caveat we need to address is that the conclusions and hypotheses proposed in this study stem from our recordings of the calcium activity of neurons. While numerous studies have used calcium imaging techniques to record neural activity, it remains unclear how accurately calcium signal levels, especially those recorded via single-photon approaches, can substitute for spiking activities. The day-to-day variability of the tracked neural ensemble and their weak activity during sleep were also major concerns since we were recording GCaMP7f activity. With major updates in the GCaMP family⁶² and the development of two-photon microscopes for freely moving animals,⁶³ these issues may be alleviated. In this study, only adult male mice were used, thus potentially limiting the generalizability of the conclusion.

embedding (t-SNE),⁵⁸ uniform manifold approximation and projection (UMAP),⁵⁹ and Isomap²⁹ have gained widespread use. Additionally, there are more advanced methods designed to integrate animal behavior with neural data, which have proven to be highly valuable.^{27,60,61} However, many studies utilizing manifold learning methods have encountered issues related to interpretability. Specifically, it can be challenging to establish direct connections between the neural manifolds and neural functions, as well as animal behavior. Our study offers an approach to circumvent this issue by tracking changes in the neural manifolds and subsequently investigating the mechanisms at the neuronal level behind these changes. As a result,

Further studies will be carried out to investigate if sex or age would impact the mechanism underlying latent learning.

The delineation of SS and WS cells warrants further investigation. In this study, we expressed Ca2 indicators in all neurons in the CA1 region and recorded their activity without categorizing cell types based on activity patterns or genetic characteristics. It is crucial to acknowledge that while the somewhat arbitrary categorization of CA1 neurons into SS and WS cells provides a valuable framework for interpreting our data, it lacks a strong biological foundation. SS cells appear to align with typical pyramidal neurons in CA1, akin to the place cells described in prior studies. However, the identity and functional

attributes of *WS* cells remain unclear due to the limitations of our approach. Previous works have shown that local interneurons can play a significant role in the process of place field development and plasticity.^{64,65} However, since *WS* cells (more than 30% of the population) significantly outnumbered the interneurons one normally finds in the local CA1 circuit, it is unlikely that *WS* cells encompass all interneurons. We hypothesize that *WS* cells include a distinct subset of pyramidal neurons that are separate from *SS* cells, which underwent plastic changes with the involvement of local interneurons. Recent investigations have shed light on the heterogeneity of pyramidal neurons in the hippocampus, revealing a broad spectrum of neural functions and synaptic connectivity among them.^{66,67} Furthermore, neurons in the pyramidal layer can be stratified by depth, with the superficial and deep layers housing neurons with distinct genetic profiles and neural functions.^{68,69} Are these two layers of neurons simply *SS* and *WS* cells? Advanced tools that enable the identification and tracking of multiple neuron types with genetic tags are imperative to delve further into this possibility.

We would like to emphasize that the bridging function of the *WS* cells should not be interpreted as evidence of direct synaptic connections among *SS* cells. There is currently insufficient evidence to support the notion that neurons in the pyramidal layer of CA1 exhibit prevalent recurrent connections.¹⁶ It is plausible that the observed changes in the neural manifold within the CA1 region represent a broader manifestation of neural plasticity spanning the entire hippocampal circuit, including its associated brain regions, such as the entorhinal circuit, thalamus, and neocortex. For instance, there is a widely accepted belief that hippocampal reactivation during sleep plays a pivotal role in memory consolidation across both the hippocampus and the cortex.^{70,71} Our observation of neural manifold changes after sleep may be an outcome of the hippocampal-cortical interactions. However, due to the technical limitation in our current setup, we were not able to accurately track animals' sleep activity or further delineate sleep stages, making our sleep data analysis more suggestive and less conclusive. In sum, these results should inspire further studies in other aspects and neural circuits involved in latent learning, deepening our understanding of the memory system.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Wei Guo (wei_guo@mit.edu).

Materials availability

This study did not generate any unique reagents.

Data and code availability

- Datasets of calcium imaging and animal behavior videos are available from the [lead contact](#) upon request.
- A list of publicly available toolboxes used for this project is included in the [key resources table](#). Custom code for data analysis is accessible on Zenodo with the linked DOIs, also listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

We thank Daniel Cho and Qianli Xu for assisting with the experiments. We thank Dr. Daniel Polley for reading and commenting on the manuscript. W.G. was funded by a PIIF VII grant. J.P.N. was funded by NIH (NRSA 1F32MH107086-01). J.J.Z. was funded by NIH (R21EY028381).

AUTHOR CONTRIBUTIONS

W.G. and M.A.W. designed the experiments; W.G. and J.J.Z. collected the data with techniques developed by J.P.N.; W.G. performed the analysis and prepared the manuscript; and M.A.W. supervised the entire project.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2024.115028>.

Received: December 14, 2023

Revised: June 26, 2024

Accepted: November 12, 2024

Published: November 28, 2024

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
pGP-AAV1-syn-jGCaMP7f-WPRE	Dana et al. ¹⁹	Addgene #104488
Experimental models: Organisms/strains		
Mouse C57BL/6J	Jax	000664
Mouse CBA/CaJ	Jax	000654
Software and algorithms		
Custom code for data processing and analysis	Custom made	https://doi.org/10.5281/zenodo.13997267
MATLAB	Mathworks	https://www.mathworks.com/
Python	Python	https://www.python.org/
Scikit-learn	scikit-learn	https://scikit-learn.org/
Caiman	Giovannucci et al. ²⁴	https://github.com/flatironinstitute/CalmAn
Bonsai	Lopes et al. ²²	https://open-ephys.org/bonsai
CellReg	Sheintuch et al. ⁴²	https://github.com/zivlab/CellReg
Other		
C&B Metabond	Parkell	S380
GRIN lens	Edmund	64–519
Loctite 454	Uline	S-15897
UCLA miniscope	Cai et al. ²¹	miniscope.org
Corrugated plastic sheets	The Home Depot	# 1000118572
KwikSil	WPI	KWIK-SIL
Ortho-Jet™ Package	Lang	Ortho-Jet&trade
Antifade Mounting Medium with DAPI	VECTASHIELD	H-1200-10
Leica CM1950	Leica	https://www.leicabiosystems.com/us/histology-equipment/cryostats/leica-cm1950/
Nikon ECLIPSE Ti	Nikon	https://www.microscope.healthcare.nikon.com/products/inverted-microscopes/eclipse-ti2-series

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

All procedures were approved by the Massachusetts Institute of Technology Committee on Animal Care and followed the guidelines established by the National Institutes of Health for the care and use of laboratory animals. A total of 16 male mice aged between 8 weeks and 20 weeks of either C57/B6 or CBA/CaJ background were used in the experiments. Animals were maintained on 12-h/12-h light/dark cycles, with the dark cycle from 9 p.m. to 9 a.m. The experiments were carried out during the light cycle, typically from 1 p.m. to 5 p.m.

METHOD DETAILS

Virus-mediated gene delivery

The mice underwent an initial anesthetization with 5% isoflurane in oxygen and were then secured in a stereotaxic surgical frame (Kopf). Throughout the procedure, the animals were maintained at a surgical plane of anesthesia with continuous 1.5%–2% Isoflurane in oxygen. Body temperature was carefully controlled and maintained at approximately 36.5°C using a homeothermic blanket system (Kent Scientific).

The surgical area was locally anesthetized with a subcutaneous injection of lidocaine (5 mg/mL). An incision was made along the midline of the scalp to expose the skull. A burr hole was then carefully created in the skull, positioned at –1.5 mm medial-lateral and 2.1 mm rostral-caudal to the bregma landmark, using a dental drill.

A motorized injector was employed to deliver 0.5 μ L of pGP-AAV1-syn-jGCaMP7f-WPRE (with a titer of approximately 1.9×10^{12} gc/ml) into the burr hole, reaching 1.5 mm beneath the skull surface at a controlled injection speed of 0.05 μ L/min. Following the completion of the injection, the pipette remained in the brain for a duration of 5 min before being carefully removed.

After this procedure, the scalp was sutured closed, and analgesic (Buprenex, 0.05 mg/kg) was administered to the animals prior to their recovery.

GRIN lens implant

One to two weeks after the virus injection, the mice were once again anesthetized to reach a surgical plane using the previously described anesthesia protocol, which included general anesthesia, local anesthesia, and body temperature control.

The surgical procedure involved the removal of the scalp and cleaning of the periosteum on the dorsal surface of the skull with 70% ethanol. The skull surface was prepared by roughening it with an etchant gel (C&B Metabond) and then thoroughly cleaned once more. A craniotomy, measuring 2 mm–2.2 mm in diameter, was created on the skull, centered on the original injection burr hole. Careful aspiration of cortical tissue exposed the rostral-caudal striation pattern located just above dorsal CA1. A GRIN lens (1.8 DIA, 0.25 pitch, 670 nm, Edmund) was meticulously placed on top of CA1 and secured to the skull with adhesive (Loctite 454). The entire exposed skull, along with the side of the GRIN lens, was covered first with clear dental cement (C&B Metabond) and then with black cement (fast-curing orthodontic acrylic resin, Lang) to block ambient light. To protect the GRIN lens, the exposed surface was covered with fast-curing silicone gel (KwikSil). Following this procedure, the animals were administered analgesic (Buprenex, 0.05 mg/kg), awakened, and allowed a minimum of two weeks for recovery.

Baseplate implant

The mice were subjected to anesthesia once more using the previously described method. A baseplate, connected to a miniscope (UCLA miniscope),²¹ was used for imaging through the GRIN lens. The angle and focus were manually adjusted to achieve an optimal field of view and image quality. The baseplate was initially secured in place with adhesive (Loctite 454) and then firmly attached to the skull using dental cement (M&B Metabond). Subsequently, the animals were allowed to recover.

Calcium imaging in freely moving animals

After allowing the animals to fully recover from the previous procedures, we initiated *in vivo* awake recording sessions. First, the animals were given time to acclimate to the miniscope and tethering, which took place over several sessions within the home cage. During this period, we also made fine adjustments to the miniscope's focus and illumination power to ensure optimal image quality. When the animals displayed normal behavior and moved without difficulty, we progressed to the maze exploration experiments.

The mazes used in the experiments were constructed from corrugated plastic sheets obtained from Home Depot. Their specific shapes and dimensions are detailed in Figure S1. To record Ca^{2+} activity from the miniscope (at a sampling rate of 30 Hz) and simultaneously monitor the animals' behavior using a webcam (at a 60 Hz sampling rate), we employed the Bonsai software package. Animal tracking was accomplished by monitoring a red tape attached to the head-mounted miniscope against the white floor background.

Sleep recording and sleep disruptions

During Ca^{2+} recordings in sessions related to sleep or quiet wakefulness, animals equipped with the miniscope were allowed to move freely within their home cage, situated within an enclosure that offered a distinct visual context from the mazes (as depicted in Figure S1). The recording process commenced once the animal exhibited prolonged immobility and typically lasted around 30 min. Continuous monitoring of the animal's movements took place throughout the recording. For recordings related to maze running after a period of sleep, we waited until the animal had fully awakened and demonstrated normal exploratory behavior.

In the case of sleep disruption experiments, the animals were placed in their home cage for a 3-h interval between two maze runs. In the sleep group, mice were allowed to rest undisturbed for the duration of this period. In contrast, the no-sleep group experienced interruptions to prevent them from falling asleep. These interruptions involved infrequent tapping on the cage or rustling the bedding, and this disturbance also spanned 3 h. The animals' movements were tracked using real-time computer vision algorithms (CV package in Bonsai). We tentatively defined the animal's state as asleep or in a state of quiet wakefulness when their locomotion speed remained below 5 cm/s for more than 0.5 min.

Histology

The animals were humanely euthanized by administering a lethal dose of Fatal Plus (100 mg/kg). The procedure involved initial perfusion with saline followed by fixation using a 10% formalin solution. The brain was subsequently extracted and post-fixed in a 10% formalin solution overnight. Afterward, the brain was transferred to a 30% sucrose solution in 0.01 M phosphate-buffered saline. Coronal sections, each 50 μ m thick, were prepared using a cryostat (Leica), stained with DAPI (Vectashield), and examined under a fluorescent microscope (Nikon Ti).

QUANTIFICATION AND STATISTICAL ANALYSIS

Ca²⁺ data analysis

The Ca²⁺ recordings underwent analysis using the CalmAn software package,²⁴ employing the CNMF-E algorithm, as outlined in. In general, the parameters we used were

- $p = 2$ # order of the autoregressive system
- $K = \text{None}$ # upper bound on number of components per patch, in general None for 1p data
- $g\text{Sig} = (3, 3)$ # gaussian width of a 2D Gaussian kernel, which approximates a neuron
- $g\text{Siz} = (13, 13)$ # average diameter of a neuron, in general $4 \times g\text{Sig} + 1$
- $\text{Ain} = \text{None}$ # possibility to seed with predetermined binary masks
- $\text{merge_thresh} = .7$ # merging threshold, max correlation allowed #0.7
- $\text{rf} = 40$ # half-size of the patches in pixels. e.g., if $\text{rf} = 40$, patches are 80×80
- $\text{stride_cnmf} = 15$ # amount of overlap between the patches in pixels (keep it at least large as $g\text{Siz}$, i.e., 4 times the neuron size $g\text{Sig}$)
- $\text{tsub} = 2$ # downsampling factor in time for initialization, increase if you have memory problems
- $\text{ssub} = 1$ # downsampling factor in space for initialization, increase if you have memory problems you can pass them here as boolean vectors
- $\text{low_rank_background} = \text{None}$ # None leaves background of each patch intact, True performs global low-rank approximation if $\text{gnb} > 0$
- $\text{gnb} = 0$ # number of background components (rank) if positive, else exact ring model with following settings; $\text{gnb} = 0$: Return background as b and W ; $\text{gnb} = -1$: Return full rank background B ; $\text{gnb} < -1$: Don't return background
- $\text{nb_patch} = 0$ # number of background components (rank) per patch if $\text{gnb} > 0$, else it is set automatically
- $\text{min_corr} = 0.8$ # min peak value from correlation image #0.8
- $\text{min_pnr} = 10$ # min peak to noise ration from PNR image #10
- $\text{ssub_B} = 2$ # additional downsampling factor in space for background
- $\text{ring_size_factor} = 1.4$ # radius of ring is $g\text{Siz} \times \text{ring_size_factor}$

Parameters specific to a few animals' recording were fine-tuned to ensure optimal extraction results. To enhance the robustness of our analysis, we excluded the initial and final 1000 frames of the recording. Subsequently, we utilized the denoised Ca²⁺ dynamics, represented by variable C within the CNMF-E model, as the activity time series for the regions of interest (ROIs).

To further refine our dataset, we removed ROIs that exhibited weak activity (indicated by a peak C value lower than 100% df/f) or sporadic activity (defined as fewer than three instances of activity exceeding 100% df/f). ROIs meeting these criteria were then selected for further analysis and tentatively regarded as neurons.

Manifold discovery

The analysis began by downsampling the recording from 30Hz to 10Hz. Subsequently, each neuron's Ca²⁺ response was normalized within the range of 0–1. This processing yielded an $N \times T$ matrix, with N representing the number of neurons and T signifying the number of frames in the recording. The Isomap algorithm from the *scipy-learn* library was employed to embed the neural data into two dimensions (three dimensions as shown in Figure S3), resulting in a $2 \times T$ matrix.

To find the optimal embedding of the neural recordings, we systematically explored different numbers of nearest neighbors (NNs) for the Isomap algorithm, ranging from 5 to 100 in increments of 5. At each increment, we compared the reconstruction error of the new embedding with the previous one. The algorithm ceased when the improvement in reconstruction error plateaued, defined as an increase of less than 10% for two consecutive NNs. Empirically, we found that this approach yielded manifolds that most closely resembled the animal's spatial experience without succumbing to overfitting (as depicted in Figures S2E–S2H).

To compute the similarity score between a neural manifold and the enclosure, it was necessary to identify the optimal angular rotation of the manifold that best matched the maze. We rotated the neural manifold incrementally in 5-degree steps (as shown in Figures S2A–S2D). For each rotated manifold, two correlation coefficients were computed: one between the manifold trajectory and the animal's physical trajectory along the x-coordinate, and the other along the y-coordinate. The similarity score of the manifold was determined by averaging these two correlation coefficients. The rotated manifold with the highest similarity score was selected as the best-matched manifold.

In the sleep data analysis, the neural manifold was calculated similarly, utilizing data exclusively from the maze run portion. The same mapping was applied to embed the recordings during sleep sessions into the neural subspace during the run. This resulted in high-dimensional data during sleep being mapped to a series of 2D points. The "jumpiness" of this data was quantified by calculating the absolute value of temporal changes in coordinates averaged over time and then averaged over the two dimensions. The distance to the manifold was first computed for individual neural states as the average distance to all the points in the manifold during the maze run. Subsequently, these values were averaged across all neural states during the sleep session.

Place field and mental field calculation

To calculate the place field of a neuron, the entire field of view, encompassing areas inaccessible to the animals, was partitioned into a 40 × 40 grid. For each grid location, the average neural response when the animal traversed that area was computed as the place field response. It adhered to the conventional practice to exclude neural responses during fidgeting or quiet wakefulness (speed <10 cm/s) in place field calculation. Similarly, the mental field of the neuron was determined using the same approach, but within the 2D neural manifold space, also divided into a 40 × 40 grid.

The information content of a neuron's place field or mental field tuning was quantified with the following formula:

$$I = \sum_i \left(\frac{\lambda_i}{\bar{\lambda}} \right) \cdot \log_2 \left(\frac{\lambda_i}{\bar{\lambda}} \right) \cdot p_i$$

And the sparsity was assessed using this formula:

$$S = \frac{(\sum p_i \lambda_i)^2}{(\sum p_i \lambda_i^2)}$$

In these equations, p_i represents the probability of the animal occupying the i th bin (occupancy time divided by total time), λ_i is the average neural response in the i th bin, and $\bar{\lambda}$ is the overall average neural response. It's essential to note that, since we used normalized neural activity in these calculations, the unit of information content was expressed in bits.⁴¹

Neurons were provisionally categorized as strongly spatial (SS) cells and weakly spatial (WS) cells based on their place field information content. The dividing threshold was chosen as the mode of the distribution, which was 0.4 bits.

Neural correlation analysis

The correlation score for a neuron was computed as the mean absolute value of the correlation coefficients between its activity and the activity of all other neurons. This measure could also be limited to calculating correlations with either SS cells or WS cells.

Cell identity tracking over sessions

For recordings obtained on the same day, we concatenated multiple data blocks and applied the CNMF-E algorithm to maintain consistent cell identities. In the case of recordings spanning multiple days, we conducted individual CNMF-E analyses for each session, and the CellReg algorithm was utilized to align the sessions based on the spatial correlations of the ROI footprints (variable A in the CNMF-E model). Only cells that consistently appeared across all learning sessions (Cell tracking quality value >0.9 in the CellReg algorithm) were included in the analysis presented in [Figures 5, 6, and 7](#).

Neural signal analysis

The neuron footprint correlation is calculated using the CalmAn neuron footprint output and the CellReg tracking results. Each session-to-session pair is calculated once and the averaged value of all pairs is designated as the mean footprint correlation of the neuron.

The peak signal strength of a neuron is the averaged value of its top-ten peak activity. The averaged signal strength of a neuron is the averaged value of its entire trace. The variability of signal of a neuron is the standard deviation of its entire trace. The signal-to-noise ratio is the peak signal strength divided by the standard deviation of a neuron's trace.

To calculate the place field size of a neurons, we first find out all pixels in the place field grid that were occupied by the animal during recording and their associated averaged neural activity levels. Then the mean and standard deviation of all pixel-related activity level was calculated. Pixels with activity exceeding two standard deviations above the mean level was defined as part of the place field. The place field size was defined as a ratio between the number of place field pixels and all occupied pixels.

Statistical analysis

All statistical analyses were conducted using MATLAB (Mathworks). Descriptive statistics are presented as mean ± SEM, unless specified otherwise. To assess the normality of the dataset, the Lilliefors test was performed when parametric statistical tests (e.g., Student's t-tests, ANOVA) were applied. Non-parametric statistical tests were employed in specific instances where the data samples did not follow a normal distribution. When the same data sample was used for multiple comparisons, the Holm-Bonferroni correction was applied to account for the increased likelihood of Type-I errors. Statistical significance was defined as $p < 0.05$. All detailed statistical results are included in the figure legends.

Cell Reports, Volume 43

Supplemental information

**Latent learning drives sleep-dependent
plasticity in distinct CA1 subpopulations**

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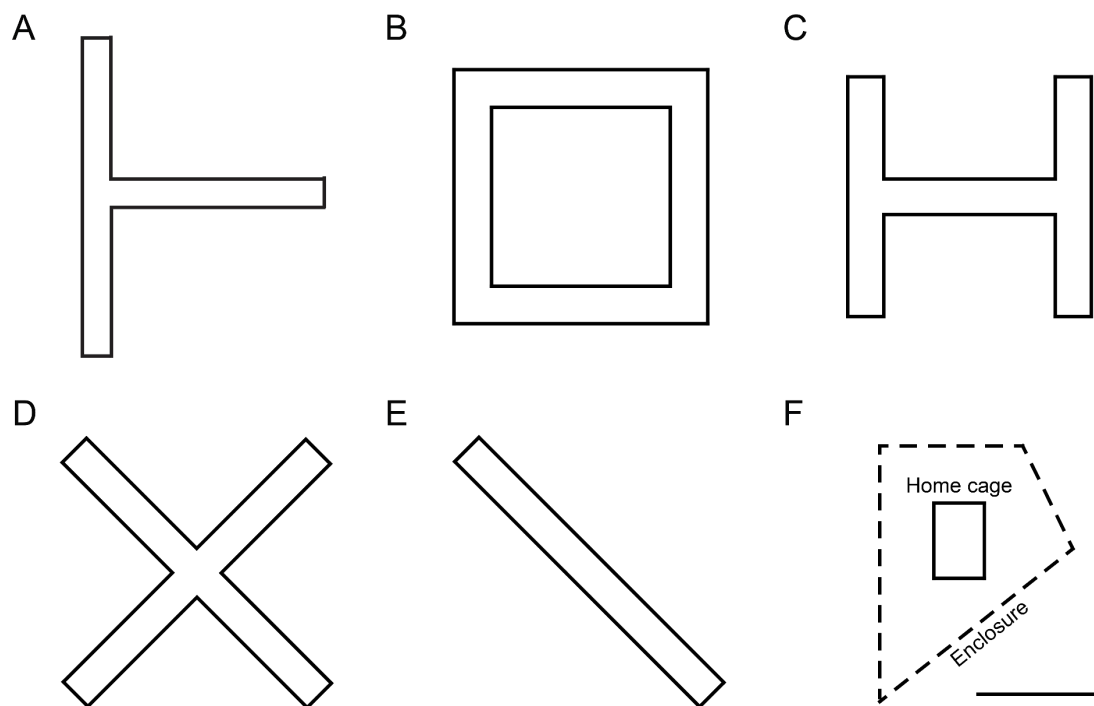


Fig. S1. Geometric shapes of mazes used in the experiments. Related to Figure 1. (A-E) Mazes. (F) Home cage placement. Scale bar: 0.5m.

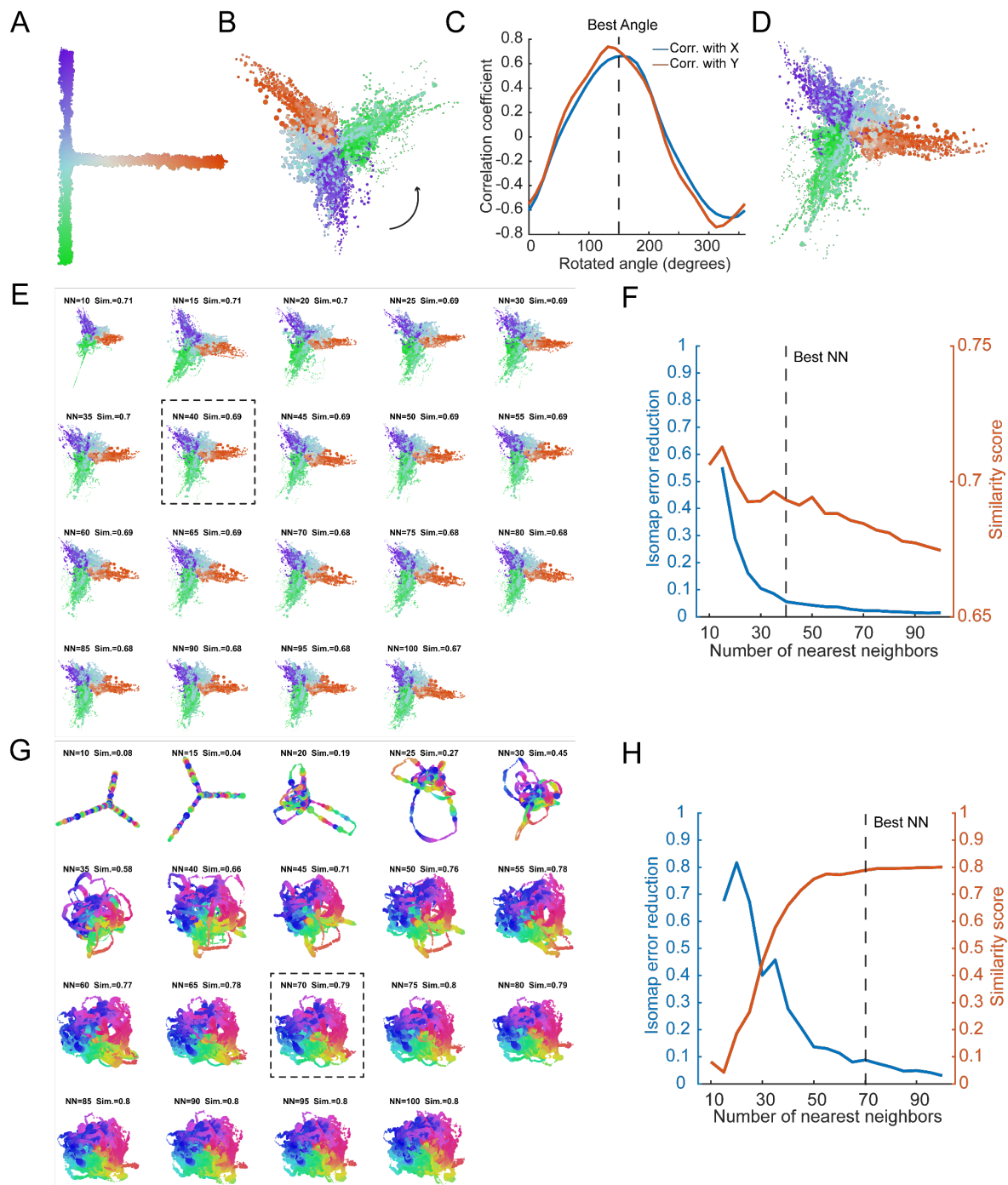


Fig. S2. Parameter optimization for manifold discovery. Related to Figure 2. (A-D) Optimal angle for manifold-maze alignment (See Methods). Neural manifold in B is rotated from 0 to 360 degrees, with the correlation coefficients with both dimensions calculated at each angle (C). The best angle is chosen as the angle associated with maximum averaged correlation coefficients (C, D). (E-F, G-H) Two examples of identifying the optimal NN for manifold discovery. Using the criterion (less than 10% reduction in embedding construction error), we can identify the manifold that optimally matches the maze (plateau in the similarity score curve, boxed manifold).

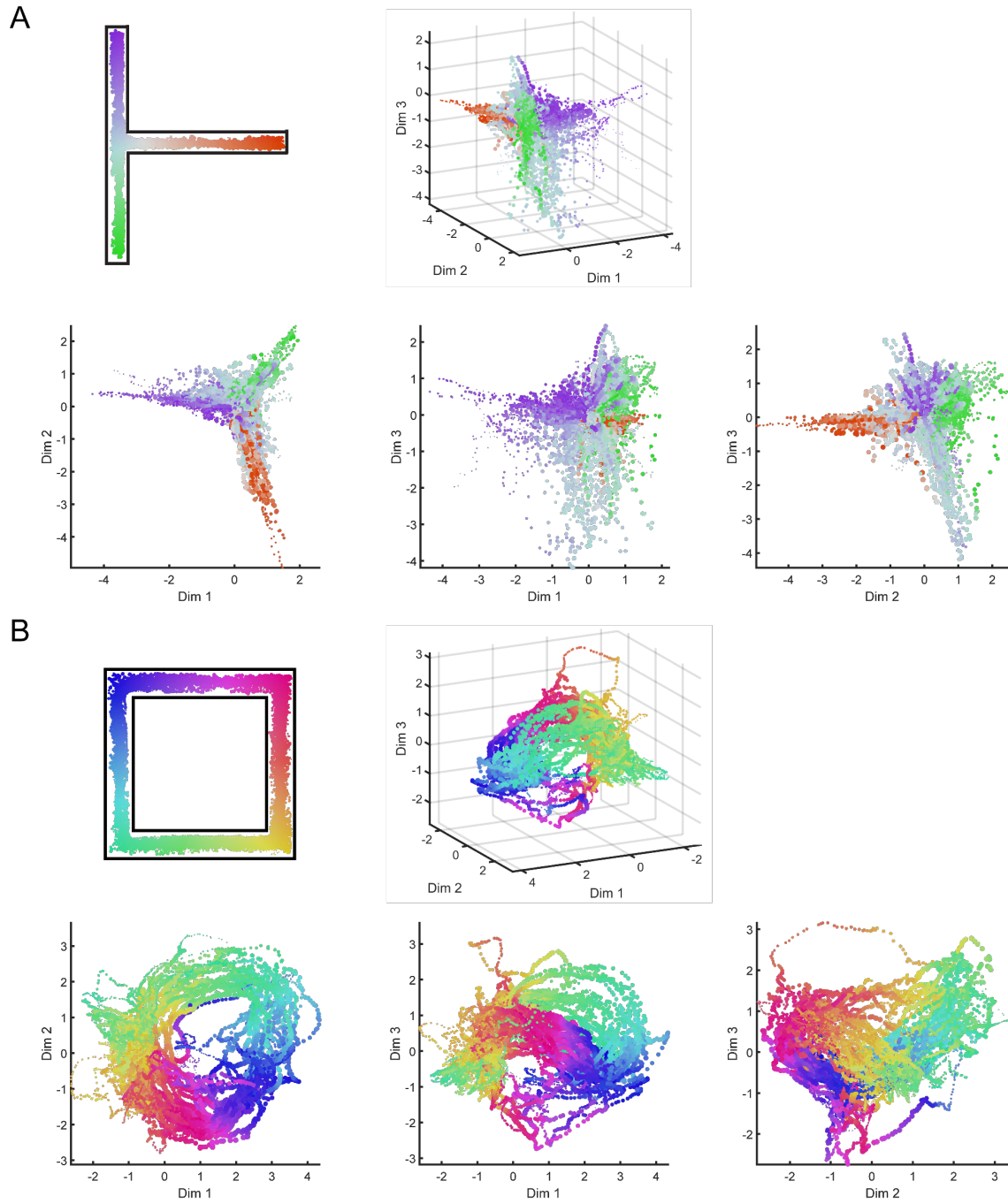


Fig. S3. Extra dimensions of the neural manifold. Related to Figure 1. (A) 3D manifold representation of data in Fig 1H. (B) 3D manifold representation of data in Fig 1I.

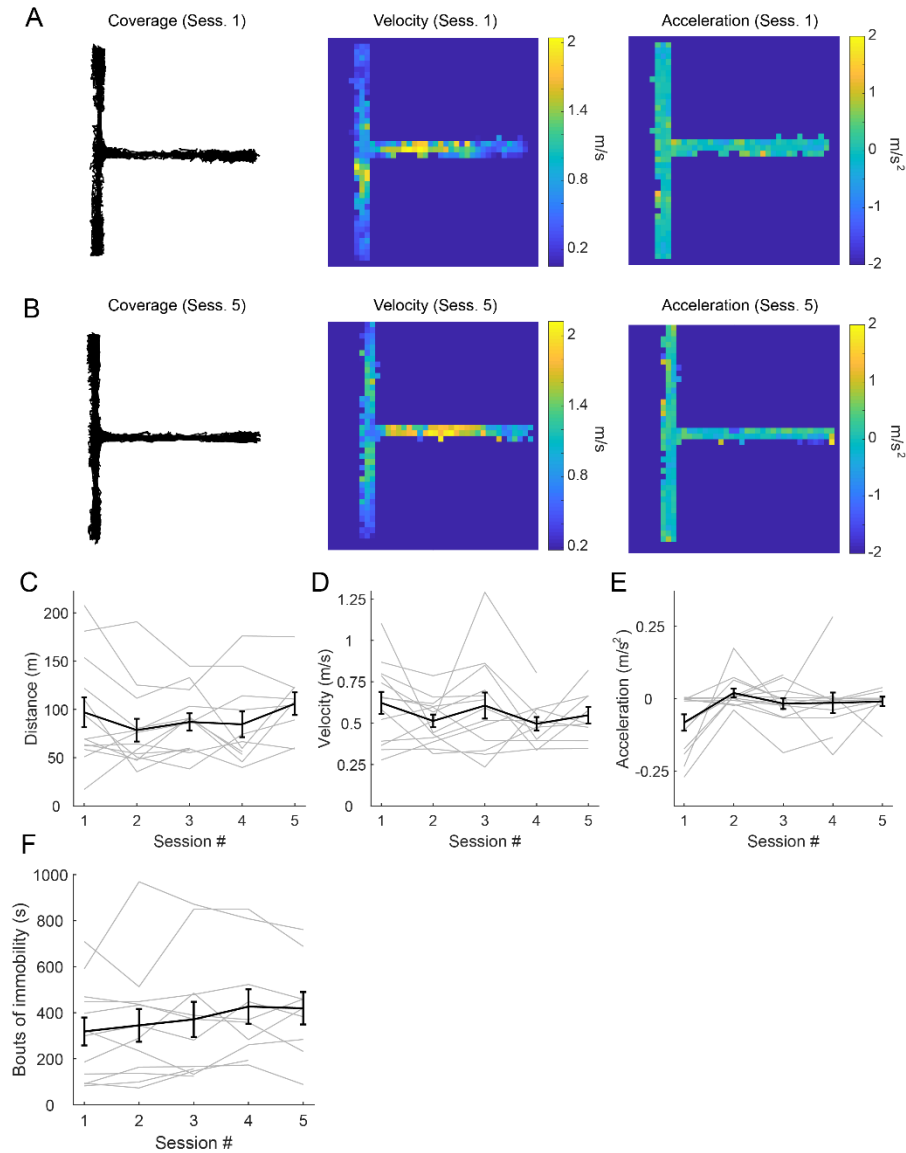


Fig. S4. Overt behavior of animals during latent learning does not change. Related to Figure 1. (A) Example coverage, velocity, and acceleration of mouse's first session exploring a maze. (B) Data from the same mouse in (A) during its fifth session in the same maze. (C) Average distance travelled of the animals during exploration (Main effect for time: $F_4 = 1.79$, $P = 0.15$, repeated measures ANOVA. 13 mazes, 10 mice). (D) Average velocity of the animals during exploration (Main effect for time: $F_4 = 1.53$, $P = 0.21$, repeated measures ANOVA. 13 mazes, 10 mice). (E) Average acceleration of the animals during exploration (Main effect for time: $F_4 = 0.94$, $P = 0.71$, repeated measures ANOVA. 13 mazes, 10 mice). (F) Average bouts of immobility (velocity Less than 5 cm/s for longer than 1s) of the animals during exploration (Main effect for time: $F_4 = 0.63$, $P = 0.64$, repeated measures ANOVA. 13 mazes, 10 mice). All error bars show mean and s.e.m. of the data.

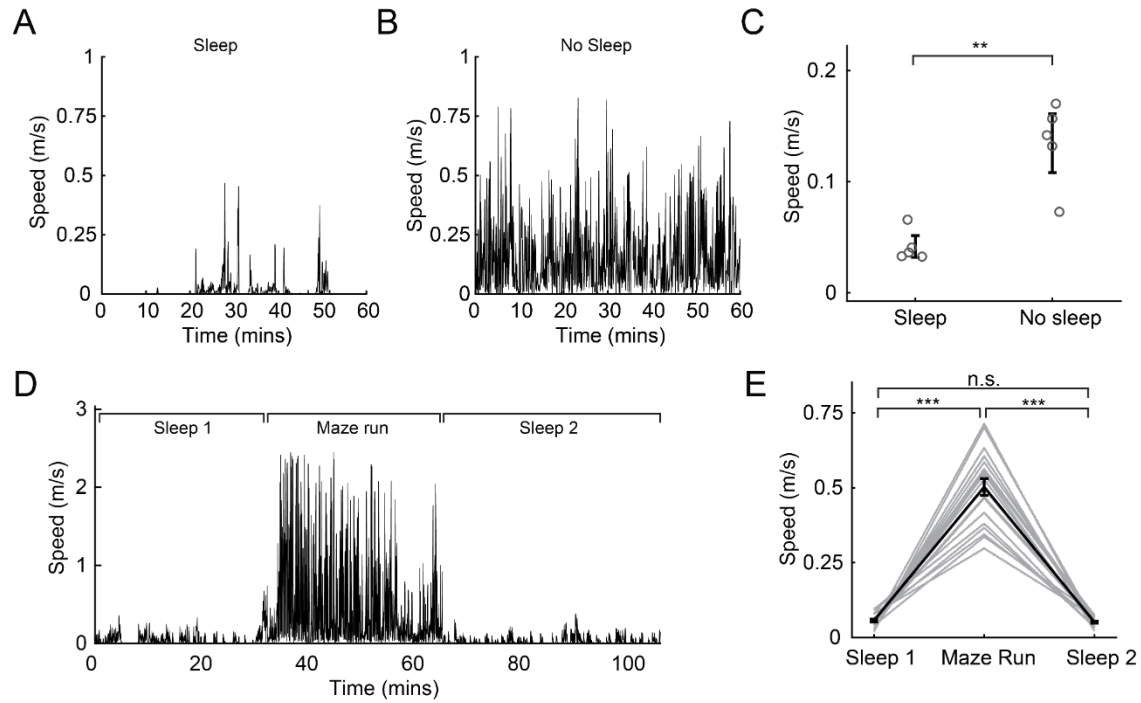


Fig. S5. Movement tracking during sleep. Related to Figure 2. (A) Example trace of animal's speed from the sleep group. (B) Example trace of animal's speed from the no-sleep group. (C) Group data showing the average speed during home cage stay was significantly different between groups ($T_4 = 6.1$, $P = 0.004$, unpaired Student's T-test. 5 sessions, 4 mice). (D) Example trace of an animal's during the sleep-run-sleep. (E) Group data showing the average speed of animals during the 3 stages of the session (Sleep 1 versus Run: $T_{18} = 14.3$, $P = 2.91 \times 10^{-11}$; Sleep 2 versus Run: $T_{18} = 15.4$, $P = 7.90 \times 10^{-12}$; Sleep 1 versus Sleep 2: $T_{18} = 0.8$, $P = 0.43$. With Holm-Bonferroni correction for multiple comparisons. 19 sessions, 4 mice). **: $P < 0.01$, ***: $P < 0.001$. All error bars show mean and s.e.m. of the data.

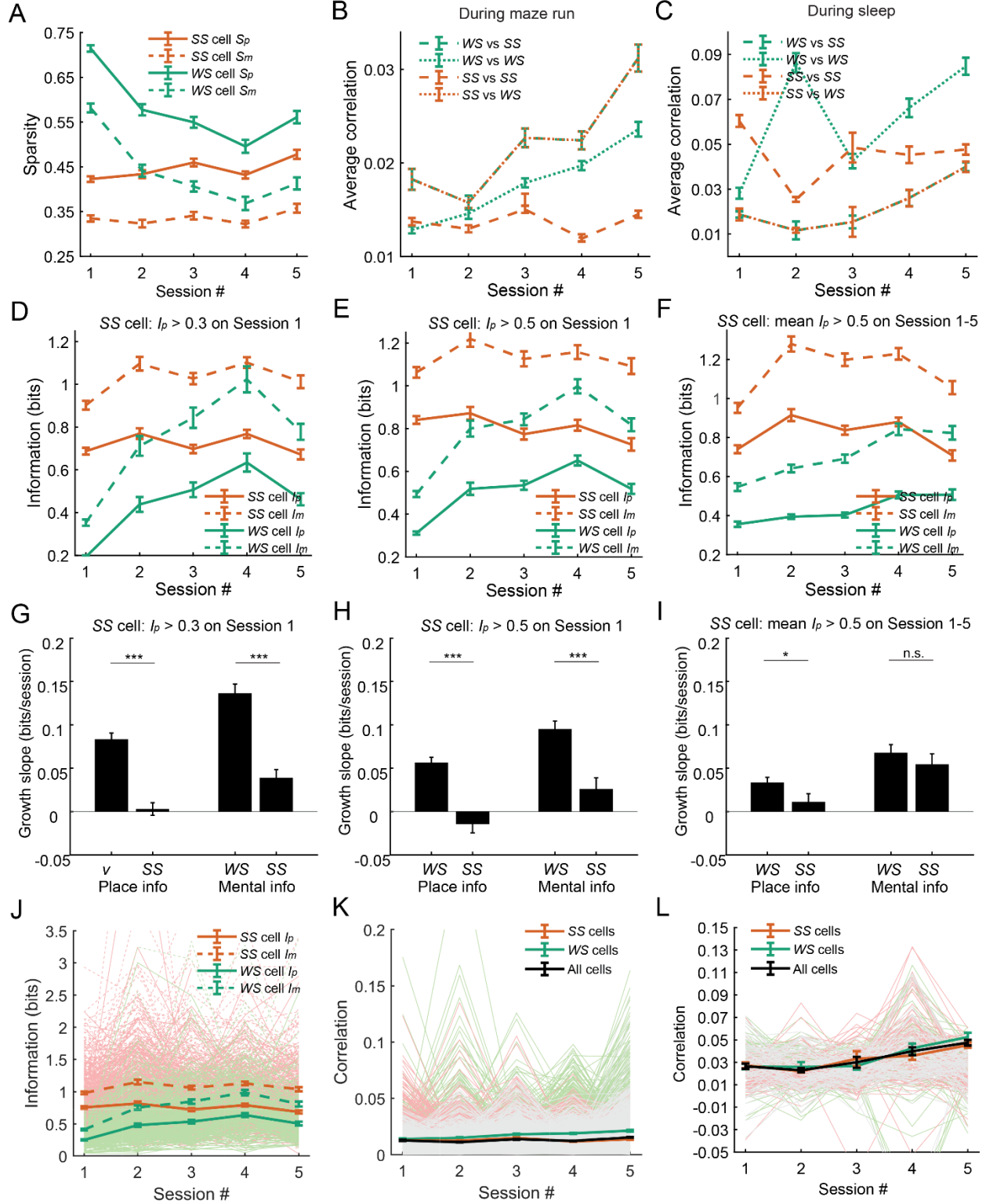


Fig. S6. Additional analyses for SS and WS cells. *Related to Figure 5-7.* (A) Sparsity changes of tracked neurons. (SS cells: main effect for time: $F_4 = 2.16$, $P = 0.07$; main effect for measurement type: $F_1 = 170.0$, $p = 2.02 \times 10^{-33}$, interaction: $F = 0.93$, $P = 0.44$. WS cells: main effect for time: $F_4 = 10.5$, $P = 2.81 \times 10^{-8}$; main effect for measurement type: $F_1 = 107.5$, $p = 4.9 \times 10^{-21}$, interaction: $F = 0.2$, $P = 0.96$. 330 SS cells and 176 WS cells from 10 mice) (B) Average correlation changes of neurons during maze run over five sessions, split into two groups: SS cells and WS cells. (WS versus SS: $F_4 = 60.0$, $P = 9.95 \times 10^{-50}$; WS versus WS: $F_4 = 27.9$, $P = 7.91 \times 10^{-23}$; SS versus SS: $F_4 = 12.82$, $P = 2.16 \times 10^{-10}$; SS versus WP: $F_4 = 11.04$, $P = 6.84 \times 10^{-9}$; All main effects for time, one-way ANOVA. 330 SS cells and 176 WS cells from 10 mice). (C) Average correlation changes of neurons during sleep over five sessions, split into two groups: SS cells and WS cells. (WS versus SS: $F_4 = 14.8$, $P = 4.20 \times 10^{-5}$; WS versus WS: $F_4 = 6.0$, $P = 4.21 \times 10^{-3}$; SS versus SS: $F_4 = 3.7$, $P = 2.5 \times 10^{-2}$; SS versus WP: $F_4 = 14.8$, $P = 4.2 \times 10^{-5}$; All main effects for time, one-way ANOVA. 257 SS cells and 216 WS cells from 4 mice). (D) Information change of tracked SS and WS cells over learning. SS cell defined as cells with $I_p > 0.3$ on session 1. (SS cells: main effect for time: $F_4 = 0.34$, $P = 0.85$; main effect for measurement type: $F_1 = 188.3$, $p = 3.98 \times 10^{-37}$, interaction: $F = 1.72$, $P = 0.14$. WS cells: main effect for time: $F_4 = 0.78$, $P = 0.53$; main effect for measurement type: $F_1 = 58.5$, $P = 2.08 \times 10^{-12}$, interaction: $F = 2.8$, $P = 0.02$. 396 SS cells, 110 WS cells from 10 mice. Repeated measures ANOVA). (E) Information change of tracked SS and WS cells over learning. SS cell defined as cells with $I_p > 0.5$ on session 1. (SS cells: main effect for time: $F_4 = 1.03$, $P = 0.39$; main effect for measurement type: $F_1 = 146.7$, $P = 1.09 \times 10^{-28}$, interaction: $F = 1.08$, $P = 0.36$. WS cells: main effect for time: $F_4 = 1.02$, $P = 0.38$; main effect for measurement type: $F_1 = 131.9$, $P = 5.01 \times 10^{-26}$, interaction: $F = 2.86$, $P = 0.02$. 257 SS cells, 249 WS cells from 10 mice. Repeated measures ANOVA). (F) Information change of tracked SS and WS cells over learning. SS cell defined as cells with $I_p > 0.5$ on average from session 1 to 5. (SS cells: main effect for time: $F_4 = 0.53$, $P = 0.77$; main effect for measurement type: $F_1 = 192.6$, $P = 2.34 \times 10^{-36}$, interaction: $F = 1.19$, $P = 0.31$. WS cells: main effect for time: $F_4 = 0.28$, $P = 0.88$; main effect for measurement type: $F_1 = 184.8$, $P = 6.79 \times 10^{-33}$, interaction: $F = 4.64$, $P = 0.001$. 295 SS cells, 211 WS cells from 10 mice. Repeated measures ANOVA). (G) Grow slope comparison among cells in panel D. (Place information: $T_{504} = 5.59$, $P = 5.10 \times 10^{-8}$, Mental information: $T_{504} = 5.06$, $P = 5.65 \times 10^{-7}$, unpaired Student's T-test) (H) Grow slope comparison among cells in panel E. (Place information: $T_{504} = 6.01$, $P = 5.10 \times 10^{-9}$, Mental information: $T_{504} = 4.32$, $P = 1.85 \times 10^{-5}$, unpaired Student's T-test) (I) Grow slope comparison among cells in panel F. (Place information: $T_{504} = 1.18$, $P = 0.05$, Mental information: $T_{504} = 0.81$, $P = 0.41$, unpaired Student's T-test) (J) Fig. 5G with individual data points. (K,L) Fig. 7B with individual data points. All error bars show mean and s.e.m. of the data.

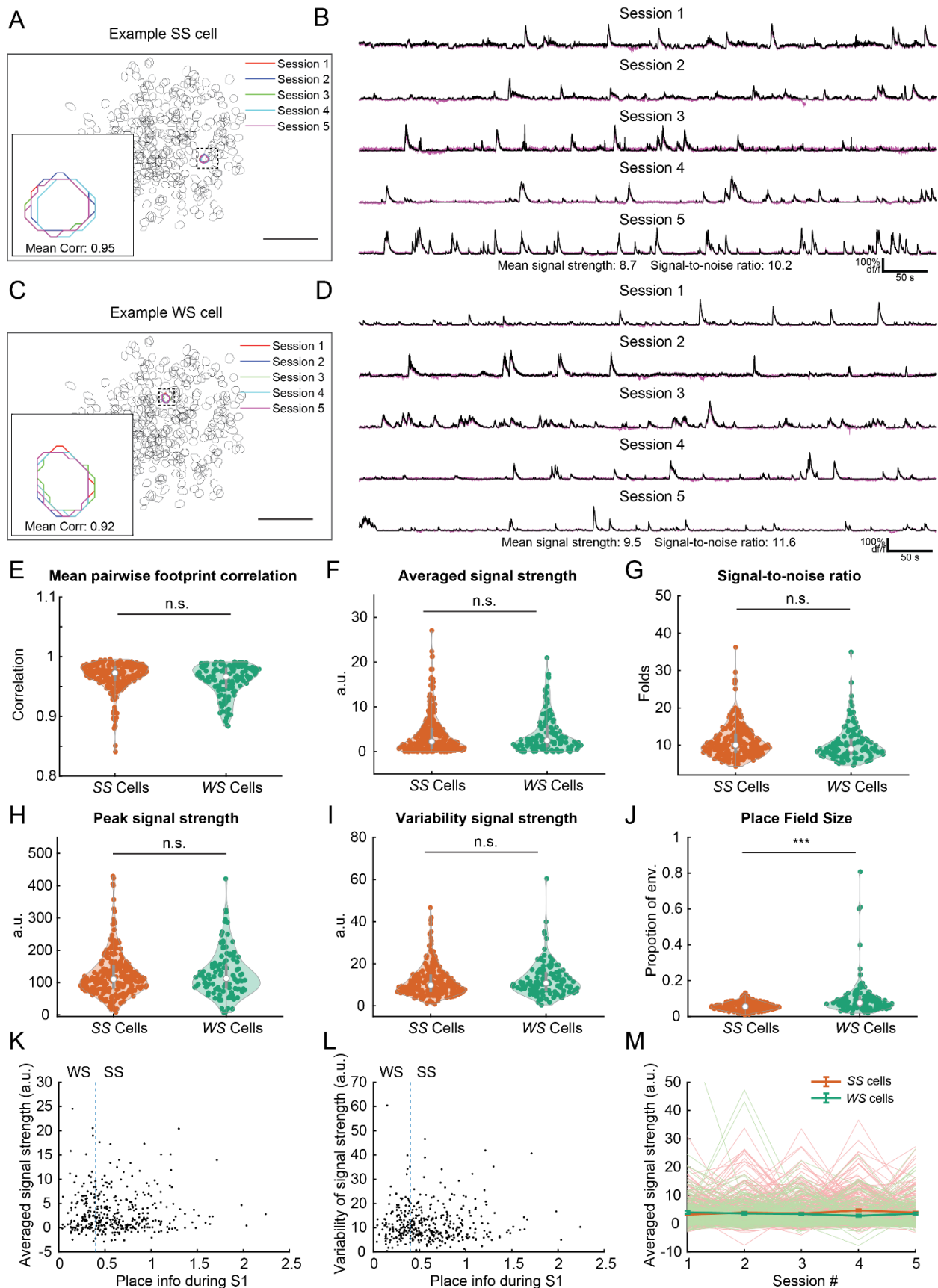


Fig. S7 Comparison of tracked SS and WS cells. Related to Figure 5-7. (A) Outlines of an example SS cell across 5 sessions. Inset: zoomed in outline of the cell. (B) Example traces of the SS cell in A across 5 sessions. Purple trace: raw ROI output; Black trace: CalmAn extraction. (C, D) Similar to (A, B) for an example WS cell. (E) Mean pairwise footprint correlation comparison between SS cells and WS cells. ($P = 0.05$, Wilcoxon's rank sum test) (F) Averaged signal strength comparison between SS cells and WS cells. ($P = 0.77$, Wilcoxon's rank sum test) (G) Signal-to-noise ratio comparison between SS cells and WS cells. ($P = 0.40$, Wilcoxon's rank sum test) (H) Peak signal strength between SS cells and WS cells. ($P = 0.69$, Wilcoxon's rank sum test) (I) Variability of signal strength comparison between SS cells and WS cells. ($P = 0.28$, Wilcoxon's rank sum test) (J) Place field size comparison between SS cells and WS cells. ($P = 1.1 \times 10^{-9}$, Wilcoxon's rank sum test) (K) Distribution of averaged signal strength of neurons against Ip during S1. (L) Distribution of s.t.d. of signal strength of neurons against Ip during S1. (M) Change of average signal strength of SS and WS neurons during all sessions. (main effect for time: $F_4 = 1.40$, $P = 0.23$; main effect for cell type: $F_1 = 8.74$, $P = 0.03$. 376 cells from 10 mice. Repeated measures ANOVA) ***: $P < 0.001$. Median value of the dataset was plotted in panels E to J. Error bars in panel M represent mean and s.e.m.